Name:		**Class:**	
Centre Number:		**Index Number:**	

DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION
Year 6

HIGHER 2 GEOGRAPHY **9173/02**

PAPER 2

Friday **20 Sept 2024** **3 hours**

Additional materials: Insert

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your Centre number, index number and name on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer Question 1 in **Section A**.
Answer Question 2 in **Section B**.
Answer **one** question in **Section C**

The number of marks is given in brackets [ ] at the end of each question or part question

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Section A

Cluster 4 Fieldwork

1 A group of six students undertook an investigation of how flood risk differs along the river near their school. The following was their research question:

"How does flood risk vary along the river near our school?"

The six students worked in pairs and each pair was assigned one of the fieldwork sites. The students collected the data on an afternoon in January. At each site, the following data was collected:

- River velocity using the float method and flowmeter
- Channel width and depth using measuring tape and rule

The students used different floats at each site. Additionally, they only had one flowmeter which was shared amongst the three pairs of students.

Resource 1A is a map showing the fieldwork sites. Resource 1B shows the land use near the river today. Resources 2A and 2B show the methods the students used to measure velocity in the river. Resources 3A and 3B shows the velocity measurements and cross-section measurement collected by the students respectively.

(a) With reference to the research question, explain if it is of a suitable scale. [3]

(b) Explain one strength and one limitation of using each method in measuring river velocity shown in Resources 2A and 2B. [4]

(c) Suggest issues related to research ethics and how the students can address them in this geographical investigation. [6]

(d) Suggest the considerations that the students should have in the selection of the investigation sites and the sampling method. [5]

(e) With reference to Resource 3A, describe the measurements recorded. [6]

(f) Suggest two limitations of the method of data representation in Resource 3B. [2]

(g) Sketch the cross-section at Site C shown in Resource 3B. [4]

(h) Evaluate the students' investigation of flood risk along the river. [10]

Section B

Cluster 3 Sustainable Future and Climate Change

2 Resource 4 shows the National Ecological footprint, Human Development Index (HDI) and urbanisation level for selected countries. Resources 5A and 5B show schematic diagrams of urban metabolic flows. Resource 6 shows the City Development Index (CDI) and its component sub-indices for selected cities in 2010.

(a) Describe the relationships between the National Ecological footprint, Human Development Index and level of urbanisation shown in Resource 4. [5]

(b) Suggest reasons for the relationship between level of urbanisation and national ecological footprint observed in Resource 4. [6]

(c) With reference to Resources 5A and 5B, explain how cities can reduce their ecological footprint and achieve sustainable urban development. [4]

(d) With reference to Resource 6, explain the usefulness and limitations of the City Development Index (CDI) in monitoring sustainable urban development. [5]

(e) Explain the impact of multiple deprivations, experienced by slum dwellers, on sustainable urban development. [5]

(f) Explain the role of human activities and associated feedback mechanisms in influencing global warming. [5]

Section C

Answer EITHER question 3 OR question 4.

3 'Urban reimaging strategies are successful in all places.'

Evaluate this statement. [20]

4 'Partnership between firms and states is important to respond to climate change successfully.'

Evaluate this statement. [20]